Supplement to "Identifying Rational Types in Unknown Environments"

Appendix C: Technical Documentation of app_DL.py and Value Net Approximators

October 13, 2026

Abstract

This supplement provides the complete technical documentation of the simulation engine implemented in `app_DL.py` and details the deep-learning-based value approximation framework described in Section 5 of the main text. It covers the file architecture and data inputs, the training scheme of the value networks, the state's dynamic optimization oracle, the simulation loop, and the assembly of the results reported in the paper.

S1 Introduction

The live execution of the Streamlit application `app_DL.py` is hosted at:

https://7kkgn4sfscykkfvv5794qw.streamlit.app/

where a dedicated "Readme" tab provides user instructions. Alternatively, the application can be run locally on both macOS and Windows systems using double-clickable launcher shortcuts; instructions for local setups are detailed in the `README.md` file in the root of the application folder. The complete research repository, including code, simulation files, and

data calibration scripts, is available at the public GitHub repository:

https://github.com/Donzhumber/TPV.git

This document outlines the application design.

Solving the dynamic mechanism's backward induction recursively is computationally prohibitive due to the need to recalculate expected continuation values. To resolve this, the model employs a value approximator based on a Multilayer Perceptron (MLP), trained in two independent instances: one approximating the continuation value that feeds the state's optimization (`captor_value_net_T10.pt`), and the other approximating the kidnapper's optimal response under its true type (`captor_true_type_value_net_T10.pt`).

This supplement details: (i) the file architecture and data inputs, (ii) the neural network configuration and the offline backward induction training scheme, (iii) the state's grid-search optimization oracle, including the dynamic belief-weighted anchoring penalty and active learning entropy subsidy, (iv) the fallback mechanism for empty feasible sets, (v) the step-by-step Bayesian simulation loop, and (vi) the assembly of the transposed Table 5.2.

S2 File Architecture

The simulation engine and deep learning models are implemented in Python. The files and their roles are summarized below:

File	Role	Status
`app_DL.py`	Streamlit application front-end; coordinates parameters, runs the simulation loop, and displays outputs and diagnostics.	Main entry point
`train_captor_value_net.py`	Defines the `CaptorValueNet` network architecture, features, and the `solve_state_problem` grid-search oracle. Handles value net training.	Imported by `app_DL.py`
`train_captor_true_type_net.py`	Implements the captor's true-type value network training under true-type hazards.	Imported by `app_DL.py`
`family_optimization.py`	Contains optimization functions for the family's best response.	Imported by `app_DL.py`
`run_period.py`	Standard game loop period-to-period simulation runner.	Imported by `app_DL.py`
`rational_behavior.py`	Core library containing utilities for outcomes, hazards, and payoffs.	Imported by `app_DL.py`

All deep learning operations are conducted using PyTorch, and simulation computations are written in vector-accelerated NumPy.

S3 Data Inputs and Prior Initialization

All calibration scenarios represent synthetic environments where the user configures the initial prior distribution μ_0 over the four captor types: common delinquency (DC), paramilitary groups (PAR), the National Liberation Army (ELN), and the Revolutionary Armed Forces of Colombia (FARC).

S3.1.0.0 Prior Initialization

In the application, the user sets the initial prior vector $\mu_0 \in \Delta(\Theta_K)$ using manual sliders for $\mu_0(\text{DC})$, $\mu_0(\text{PAR})$, and $\mu_0(\text{ELN})$. The prior for $\mu_0(\text{FARC})$ is then computed automatically

to satisfy the normalization constraint:

$$\mu_0(\text{FARC}) = 1.0 - \sum_{\theta \in \{\text{DC, PAR, ELN}\}} \mu_0(\theta) \tag{S3.1}$$

This allows setting the exact scenarios analyzed in the calibration sections of the paper, such as:

- **ELN Baseline Prior Scenario**: $\mu_0 = (0.25, 0.35, 0.05, 0.35)$
- **PAR Baseline Prior Scenario**: $\mu_0 = (0.55, 0.05, 0.35, 0.05)$

S3.2.0.0 Covariates and Environmental Inputs

The interface allows the user to select the following qualitative covariates to configure the environment's empirical hazard levels and structural responses:

1. **Perpetrator Group (θ_K)**: Selects FARC, ELN, PAR, or DC to index baseline technology hazards $\beta_{K,j}$.
2. **Geographic Zone (z)**: Selects Metropolis, Andean, Caribbean, Pacific / Red Zone, or Eastern Plains/Jungle to index the geographical parameters β_z, where: Metropolis $= 0.00$, Andean $= -0.45$, Caribbean $= -0.70$, Pacific/Red Zone $= -0.20$, and Eastern Plains/Jungle $= -0.32$.
3. **Victim Profile (θ_V)**: Selects Private or Public sector. A public sector victim subtracts $\beta_{V,1} = 1.36$ from the payment hazard rate index.
4. **Family Wealth (θ_F)**: Selects Standard or High wealth class. A high-wealth family adds $\beta_{F,1} = 0.80$ to the payment hazard rate index.
5. **State Type (θ_S)**: Selects Strict or Lax. A lax state adds $\beta_{S,1} = 0.50$ and $\beta_{S,2} = 0.50$ to the payment and death hazard indices respectively.

S3.3.0.0 Structural Hyperparameters

The interface allows adjusting structural parameters such as the time horizon (T_{horizon}), active exploration weight (ψ_H), and minimum MDG temperature ($\underline{c}$), alongside calibration parameters for α_0, γ_0, T_{mad}, λ_4, and η_{cal}, which parameterize the environment's hazards and temperature decay.

S4 Deep Learning Value Network Architecture

The computational cost of solving the game's backward induction is dominated by the recursive estimation of the kidnapper's expected continuation value. To avoid recursive calculations in real time, the app leverages pre-trained networks of class `CaptorValueNet` (inheriting from `torch.nn.Module`).

S4.1.0.0 Network Layout

The neural network layout consists of a Multi-Layer Perceptron (MLP) with the following structure:

$$\text{Input (34)} \xrightarrow{\text{Linear+Tanh}} 96 \xrightarrow{\text{Linear+Tanh}} 96 \xrightarrow{\text{Linear}} 1 \text{ (Continuation Value)} \tag{S4.1}$$

The input feature vector $X \in \mathbb{R}^{34}$ encodes:

1. Belief state vector μ (4 dimensions: DC, PAR, ELN, FARC).
2. Normalized time index τ/T (1 dimension).
3. One-hot encoding of the captor type θ_K (4 dimensions).
4. Normalized ransom demand scale $R/100$ (1 dimension).
5. Payback discount rates $\tilde{\beta}$ (4 dimensions).

6. One-hot encoding of region z (5 dimensions).

7. One-hot encoding of family wealth class (2 dimensions).

8. One-hot encoding of victim profile (2 dimensions).

9. One-hot encoding of state type (2 dimensions).

10. Structural parameters: normalized $T_{\text{mad}}/30$, $T_0/2.0$, $\eta_{\text{cal}}/0.20$, $c_{\text{bar}}/0.20$, and H_{ratio} (5 dimensions).

11. Hazard variables: normalized $\lambda_4/0.01$, $\eta_1/3.0$, $\eta_2/3.0$, and $\beta_R/10.0$ (4 dimensions).

S4.2.0.0 Training Procedure

The value network is trained offline via backward induction from $\tau = 10 \to 1$:

1. For each level τ, $2,000$ state configurations μ and covariate sets are generated using Monte Carlo sampling.

2. The state's optimization problem is solved using the value net predicted for step $\tau + 1$ (or a terminal condition of $V_{T_{\max}+1} \equiv 0$).

3. The resulting Bellman value $V_\tau = \max\{U_{\text{rel}}, U_{\text{kill}}, V_{\text{cont}}\}$ is recorded as the regression target.

4. The network parameters are optimized to minimize the Mean Squared Error (MSE) using the Adam optimizer over 200 epochs per period.

Two distinct network instances are trained:

- `captor_value_net_T10.pt`: Approximates the belief-weighted expected continuation value used to solve the State's optimization problem.

- `captor_true_type_value_net_T10.pt`: Approximates the true-type specific continuation value to evaluate the kidnapper's true response.

S5 State Optimization Grid-Search Oracle

In every period τ, the state determines the optimal interdiction and pressure instruments $(\alpha_\tau^*, \gamma_\tau^*)$ that minimize social cost, subject to incentive and participation constraints. This is solved via a 101×101 grid search over $(\alpha, \gamma) \in [0,1]^2$.

S5.1.0.0 Objective Functions

The state evaluates the objective function for each branch $b \in \{\text{Rescue}, \text{Negotiate}\}$:

$$V_\tau^b(\mu_\tau) = \min_{(\alpha,\gamma)\in[0,1]^2} \left\{ \sum_\theta \mu_\tau(\theta) L_\tau^b(\theta, \alpha, \gamma) + \Pi_b^S(\alpha, \gamma; \mu_\tau) - \psi_H \Delta H_\tau(\alpha, \gamma) \right\} \tag{S5.1}$$

where:

1. $L_\tau^b(\theta, \alpha, \gamma)$ is the structural social loss of the branch:

$$L_\tau^{\text{Rescue}}(\theta, \alpha, \gamma) = \omega_k(1 - p_{\text{surv}}(\theta, \theta)) + C_{\text{ops}}(\gamma, \alpha; \theta) \tag{S5.2}$$

$$L_\tau^{\text{Negotiate}}(\theta, \alpha, \gamma) = \omega_p R(1 - \alpha) + \omega_k \bar{h}_2(\theta, \alpha, \gamma) + C_{\text{maint}}(\gamma, \alpha; \theta) \tag{S5.3}$$

   with C_{ops} and C_{maint} representing quadratic operational and maintenance costs.

2. $\Pi_b^S(\alpha, \gamma; \mu_\tau)$ is the quadratic anchoring penalty (deviation premium):

$$\Pi_b^S(\alpha, \gamma; \mu_\tau) = \chi_\alpha \left(\alpha - \alpha_b^\mu(\mu_\tau)\right)^2 + \chi_\gamma \left(\gamma - \gamma_b^\mu(\mu_\tau)\right)^2 \tag{S5.4}$$

   which penalizes deviations from the belief-weighted certainty benchmarks $\alpha_b^\mu(\mu_\tau) = \sum_\theta \mu_\tau(\theta) \alpha_b^{\theta,*}$ and $\gamma_b^\mu(\mu_\tau) = \sum_\theta \mu_\tau(\theta) \gamma_b^{\theta,*}$.

3. $-\psi_H \Delta H_\tau(\alpha, \gamma)$ is the active exploration subsidy, where ΔH_τ is the expected reduction in prior entropy resulting from the Bayes update.

S5.2.0.0 Constraints

The optimization is masked over the feasible grid space $\Gamma_\tau(\mu_\tau) = IC^K \wedge IR^K \wedge IR^F$:

1. **Captor Incentive Compatibility (IC^K)**:

$$\min_{\theta_j} \sum_{\theta_i} \mu_\tau(\theta_i) \left[U_b(\theta_i) - U_{b(\theta_j)}(\theta_i) \right] \geq 0 \tag{S5.5}$$

2. **Captor Participation (IR^K)**:

$$\sum_{\theta} \mu_\tau(\theta) \left[U_{\text{rel}}(\theta) - \max\{V_{\text{cont}}(\theta), U_{\text{kill}}(\theta)\} \right] \geq 0 \tag{S5.6}$$

3. **Family Participation (IR^F)**:

$$U_{\text{coop}}(\theta_F) \geq U_{\text{col}}(\theta_F) \tag{S5.7}$$

with:

$$U_{\text{coop}}(\theta_F) = \mathbb{E}_\mu[p_{\text{surv}}] V_F^L - e_{\tau,F}(\gamma) \tag{S5.8}$$

$$U_{\text{col}}(\theta_F) = \mathbb{E}_\mu[p_{\text{rel}}] V_F^L - R - \mathbb{E}_\mu[p_{\text{det}}] F_{\text{col}} \tag{S5.9}$$

where V_F^L is the family's valuation of the victim's life ($V_F^L = 200.0$), F_{col} is the legal penalty/fine for private collusion ($F_{\text{col}} = 20.0$), and $e_{\tau,F}(\gamma) = \Phi_F(\theta_F) e^{\kappa_F(\theta_F)\gamma} + \nu_F(\theta_F)$ represents the family's operational cost of cooperating with GAULA interdiction.

S6 Fallback Mechanism

If no point on the 101×101 grid satisfies the constraints simultaneously ($\Gamma_\tau(\mu_\tau) = \emptyset$), the oracle implements a fallback protocol to prevent mathematical collapse:

1. The constraints are temporarily bypassed.

2. The objective values V_τ^{Rescue} and $V_\tau^{\text{Negotiate}}$ are computed over the entire grid.

3. The branch that yields the lower unconstrained minimum is chosen, setting the controls $(\alpha_\tau^*, \gamma_\tau^*)$ to the argmin of that branch.

4. The oracle returns the flag `feasible` set to `False`, which is recorded in the output DataFrame for diagnostic tracking.

S7 Bayesian Simulation Loop and Table 5.2 Assembly

Once initialized, the game runs period-by-period for $t = 1 \dots T_{\text{horizon}}$. In each step, the engine executes:

1. **Maturity Filter**: Computes $M(t) = \min\{1, (t/T_{\text{mad}})^2\}$, scaling hazard rates.

2. **MDG Temperature**: Updates the temperature:

$$T_t = T_0 \max\left\{\left(\frac{H(\mu_t)}{H(\mu_0)}\right) e^{-\eta_{\text{cal}} t}, \underline{c}\right\}$$

3. **Policy Inference**: Evaluates the value network to solve the state's optimization problem via `solve_state_problem`, yielding (α_t^*, γ_t^*).

4. **Intentions**: Finds the best responses a_K^* (captor) and a_F^* (family) given the policy.

5. **Frictional Draw (MDG)**: Samples executed actions $\tilde{a}_S, \tilde{a}_K, \tilde{a}_F$ via the logit frictional layer.

6. **Hazards and Absorption**: Computes Cox survival probabilities and draws whether the case continues or is absorbed (Rescue, Payment, Death, Escape, or Release).

7. **Bayesian Update**: Updates beliefs based on the observed outcome:

$$\mu_{t+1}(\theta) = \frac{\mu_t(\theta)\text{lik}_t(\theta)}{\sum_{\theta'} \mu_t(\theta')\text{lik}_t(\theta')} \tag{S7.1}$$

The resulting log is formatted into a DataFrame, rounded to 3 decimal places, and transposed so that variables are rows and periods are columns. The final output displays the first 15 columns representing periods $t = 0 \dots 14$.

S8 Calibrated Coefficients and Utility Parameters

This section presents the exact numeric values of the structural calibration parameters and coefficients used in the hazard rates and utility functions across all perpetrator types (DC, PAR, ELN, FARC) and environments, as implemented in `app_DL.py` and its backend libraries.

S8.1.0.0 Hazard Rate Functional Forms and Coefficients

The effective hazard rates $\tilde{\lambda}_j(t \mid \mathcal{C}_t)$ for the three main outcome branches (Payment, Death, and Rescue, indexed by $j \in \{1, 2, 3\}$ respectively) are defined dynamically as:

$$\begin{aligned}
\tilde{\lambda}_1(t \mid \mathcal{C}_t) &= M(t)\lambda_{10} \exp\Big(\beta_{K,1}(\theta_K) + \beta_{z,1}(z) + \beta_{V,1}(\theta_V) + \beta_{S,1}(\theta_S) \\
&\quad + \phi_{F,1}(\tilde{a}_F) + \phi_{K,1}(\tilde{a}_K) - \zeta_\alpha(\theta_K)\alpha_t - \zeta_\gamma(\theta_K)\gamma_t - z_d p_{\mathrm{det},t} \Big) \qquad \text{(S8.1)} \\
\tilde{\lambda}_2(t \mid \mathcal{C}_t) &= M(t)\lambda_{20} \exp\Big(\beta_{K,2}(\theta_K) + \beta_{z,2}(z) + \beta_{S,2}(\theta_S) - \phi_{F,2}(\tilde{a}_F) \\
&\quad + \phi^{\mathrm{kill}}_{K,2}(\tilde{a}_K) + \phi^{\mathrm{cont}}_{K,2}(\tilde{a}_K) + \zeta_\alpha(\theta_K)\alpha_t + \zeta_\gamma(\theta_K)\gamma_t - z_d p_{\mathrm{det},t} \Big) \qquad \text{(S8.2)} \\
\tilde{\lambda}_3(t \mid \mathcal{C}_t) &= M(t)\lambda_{30} \exp\Big(-\beta_{S,3}(\theta_S) + \beta_{K,3}(\theta_K) + \beta_{z,3}(z) + \zeta_\alpha(\theta_K)\alpha_t \\
&\quad + \zeta_\gamma(\theta_K)\gamma_t + z_d p_{\mathrm{det},t} + \zeta_{R,3}(\tilde{a}_S) - \phi_{F,3}(\tilde{a}_F) + \phi_{K,3}(\tilde{a}_K) \Big) \qquad \text{(S8.3)}
\end{aligned}$$

where:

- $M(t) = \min\left\{1, \left(\frac{t}{T_{\mathrm{mad}}}\right)^2\right\}$ where $T_{\mathrm{mad}} = 5.0$ is the logistic maturation filter.
- $\tilde{\lambda}_4(t) = \lambda_4 = 0.0005$ is the exogenous release hazard (constant).

- $z_d = 0.18$ represents the sensitivity to the collusion detection probability.

The baseline hazard rates λ_{j0} (scaling constants) are:

$$\lambda_{10} = 0.012, \qquad \lambda_{20} = 0.002, \qquad \lambda_{30} = 0.008$$

The action-contingent indicators ϕ represent shift parameters that activate under specific choices of family ($\tilde{a}_F \in \{\text{Pay}, \text{Refuse}\}$), captor ($\tilde{a}_K \in \{\text{Continue}, \text{Kill}, \text{Release}\}$), and state ($\tilde{a}_S \in \{\text{Rescue}, \text{Negotiate}\}$):

- $\phi_{F,1} = 3.20$ if family pays ($\tilde{a}_F = \text{Pay}$), 0 otherwise.
- $\phi_{K,1} = -1.15$ if captor continues ($\tilde{a}_K = \text{Continue}$), 0 otherwise.
- $\phi_{F,2} = -1.50$ (so the term $-\phi_{F,2} = +1.50$ when family pays), 0 otherwise.
- $\phi_{K,2}^{\text{kill}} = 4.00$ if captor kills ($\tilde{a}_K = \text{Kill}$), 0 otherwise.
- $\phi_{K,2}^{\text{cont}} = 0.50$ if captor continues, 0 otherwise.
- $\zeta_{R,3} = 2.50$ if state rescues ($\tilde{a}_S = \text{Rescue}$), 0 otherwise.
- $\phi_{F,3} = -1.00$ (so $-\phi_{F,3} = +1.00$ when family pays), 0 otherwise.
- $\phi_{K,3} = 0.50$ if captor continues, 0 otherwise.

The baseline covariate indicators represent environmental and capability factors:

- $\beta_{V,1} = 1.36$ if victim is Public Sector, 0 otherwise.
- $\beta_{F,1} = 0.80$ if wealth is High, 0 otherwise.
- $\beta_{S,1} = \beta_{S,2} = \beta_{S,3} = 0.50$ if State Type is Lax, 0 otherwise.

Table S8.1 shows the perpetrator-specific hazard offsets $\beta_{K,j}$:

Table S8.2 summarizes the geographic zone parameters β_z:

Table S8.1: Captor-Type Specific Hazard Coefficients ($\beta_{K,j}$)

Perpetrator Group	Ransom ($\beta_{K,1}$)	Death ($\beta_{K,2}$)	Rescue ($\beta_{K,3}$)
FARC	-0.70	-0.85	0.90
ELN	1.10	0.20	-0.65
PAR	-0.25	1.35	0.15
DC	1.55	-0.40	-0.95

Table S8.2: Geographic Zone Coefficients (β_z)

Geographic Zone (z)	Coefficient (β_z)
Metropolis	0.00
Andean	-0.45
Caribbean	-0.70
Pacific / Red Zone	-0.20
Eastern Plains/Jungle	-0.32

Table S8.3: Policy Sensitivity Coefficients ($\zeta_\alpha, \zeta_\gamma$)

Type (θ_K)	Financial Blocking (ζ_α)	Operational Pressure (ζ_γ)
DC	0.2409	0.5450
PAR	0.2183	0.5848
ELN	0.2101	0.5532
FARC	0.2087	0.6197

Table S8.3 shows the sensitivity coefficients $(\zeta_\alpha, \zeta_\gamma)$ that multiply the policies (α_t, γ_t):

The collusion detection probability index parameters are η_0 (DC: -1.5, PAR: -2.0, ELN: -2.5, FARC: -2.8), $\eta_1 = 1.0$ and $\eta_2 = 1.0$ (used in $p_{\text{det},t} = [1 + \exp(-(\eta_0 + \eta_1\alpha_t + \eta_2\gamma_t))]^{-1}$). The rescue lethality indices are α_{leth} (DC: -5.25, PAR: -5.15, ELN: -5.05, FARC: -4.95) and $\beta_R = 7.0$ (used in $p_{\text{surv},t} = [1 + \exp(-(\alpha_{\text{leth}} + \beta_R\iota_t))]^{-1}$).

S8.2.0.0 Utility and Payoff Function Parameters

S8.2.0.1 State's Objective Function The state chooses the action a_τ^S (Rescue or Negotiate) and the policy instruments $(\alpha_\tau, \gamma_\tau)$ to minimize the expected social loss:

$$V_\tau^S(a_\tau^S, \alpha_\tau, \gamma_\tau \mid \mu_\tau) = \sum_\theta \mu_\tau(\theta) L_\tau(a_\tau^S, \alpha_\tau, \gamma_\tau, \theta) + \Pi^S(\alpha_\tau, \gamma_\tau \mid \mu_\tau) - \psi_H \Delta H_\tau(\alpha_\tau, \gamma_\tau) \quad \text{(S8.4)}$$

where the branch-specific losses are:

$$L_\tau^{\text{Rescue}}(\theta, \alpha, \gamma) = \omega_k(1 - p_{\text{surv}}(\theta, \hat{\theta})) + C_{\text{ops}}(\gamma, \alpha) \tag{S8.5}$$

$$L_\tau^{\text{Negotiate}}(\theta, \alpha, \gamma) = \omega_p R(1 - \alpha) + \omega_k \bar{h}_2(\theta, \alpha, \gamma) + C_{\text{maint}}(\gamma, \alpha) \tag{S8.6}$$

with a baseline ransom amount $R = 100.0$.

The policy deviation anchoring penalty is defined as:

$$\Pi^S(\alpha, \gamma \mid \mu_\tau) = \chi_\alpha \left(\alpha - \sum_\theta \mu_\tau(\theta)\alpha_b^{\theta,*}\right)^2 + \chi_\gamma \left(\gamma - \sum_\theta \mu_\tau(\theta)\gamma_b^{\theta,*}\right)^2 \tag{S8.7}$$

with baseline target benchmarks $\alpha_b^{\theta,*} = 0.8$ and $\gamma_b^{\theta,*} = 0.6$. The information gain is $\Delta H_\tau(\alpha_\tau, \gamma_\tau) = H(\mu_\tau) - \mathbb{E}[H(\mu_{\tau+1}) \mid \alpha_\tau, \gamma_\tau]$.

The state's weights for social loss are:

- Ransom transfer weight: $\omega_p = 0.15$
- Lethality / Loss of life weight: $\omega_k = 200.0$
- Active learning entropy subsidy weight: $\psi_H = 25.0$
- State policy deviation anchoring weights: $\chi_\alpha = 0.8$, $\chi_\gamma = 0.5$

Operational and maintenance costs are parameterized as:

$$C_{\text{ops}}(\gamma, \alpha) = c_0 + c_1\gamma + \frac{1}{2}c_2\gamma^2 + c_3\alpha + \frac{1}{2}c_4\alpha^2 + c_5\gamma\alpha \tag{S8.8}$$

$$C_{\text{maint}}(\gamma, \alpha) = m_0 + m_1\gamma + \frac{1}{2}m_2\gamma^2 + m_3\alpha + \frac{1}{2}m_4\alpha^2 + m_5\gamma\alpha \tag{S8.9}$$

with coefficients shown in Table S8.4:

Table S8.4: State Cost Coefficients (c_k and m_k)

Type	Cost Function	c_0, m_0	c_1, m_1	c_2, m_2	c_3, m_3	c_4, m_4	c_5, m_5
DC	C_{ops}	1.00	−0.25	1.00	−0.20	1.00	0.10
	C_{maint}	3.00	−6.00	10.00	−0.80	6.00	0.10
PAR	C_{ops}	1.00	−0.40	1.00	−0.35	1.00	0.10
	C_{maint}	3.00	−6.50	10.00	−1.10	6.00	0.10
ELN	C_{ops}	1.00	−0.55	1.00	−0.50	1.00	0.10
	C_{maint}	3.00	−7.00	10.00	−1.40	6.00	0.10
FARC	C_{ops}	1.00	−0.70	1.00	−0.65	1.00	0.10
	C_{maint}	3.00	−7.50	10.00	−1.70	6.00	0.10

S8.2.0.2 Family's Payoff Function The family decides between cooperating with the state ($a_F = \text{Cooperate}$) and colluding/paying ransom ($a_F = \text{Collude}$):

$$U^F_{\text{coop},\tau}(\theta_F) = \left(\sum_{\theta \in \Theta_K} \mu_\tau(\theta)\, p_{\text{surv},\tau}(\theta) \right) V_L - e_\tau(\gamma_\tau, \theta_F) \tag{S8.10}$$

$$U^F_{\text{col},\tau}(\theta_F) = \left(\sum_{\theta \in \Theta_K} \mu_\tau(\theta)\, p_{\text{rel},\tau}(\theta) \right) V_L - R - p_{\text{det},\tau}\, F_{\text{col}} \tag{S8.11}$$

where the valuation of the victim's life is $V_L = 200.0$, the legal penalty for private collusion is $F_{\text{col}} = 20.0$, and the family's institutional cost of cooperation is $e_\tau(\gamma_\tau) = \phi_F \exp(\kappa_F \gamma_\tau) + \nu_F$, where the coefficients $(\phi_F, \kappa_F, \nu_F)$ are:

- **High wealth**: $\phi_F = 0.0600$, $\kappa_F = 3.6000$, $\nu_F = 0.0200$
- **Low/Standard wealth**: $\phi_F = 0.0600$, $\kappa_F = 3.0000$, $\nu_F = 0.0200$

S8.2.0.3 Captor's Payoff Function The captor decides whether to release ($a_K = \text{Release}$), kill ($a_K = \text{Kill}$), or continue sustainment ($a_K = \text{Continue}$):

$$V^K_\tau(\theta_K) = \max \left\{ U^K_{\text{rel},\tau}(\theta_K),\ \ U^K_{\text{kill},\tau}(\theta_K, \theta_S),\ \ V^K_{\text{cont},\tau}(\theta_K) \right\} \tag{S8.12}$$

where:

$$U^K_{\text{rel}}(\theta_K) = -\kappa_{\text{rel}}(\theta_K) \tag{S8.13}$$

$$U^K_{\text{kill}}(\theta_K, \theta_S) = \Big(1 - p_{\text{cap},\tau}\Big)\,\eta(\theta_K) - p_{\text{cap},\tau}\,F_{\text{cap}}(\theta_K, \theta_S) \tag{S8.14}$$

$$\begin{aligned} V^K_{\text{cont},\tau}(\theta_K) = {} & p_{\text{pay},\tau} R(1-\alpha_\tau) - C_\tau(\gamma_\tau, \theta_K) - p_{\text{cap},\tau} F_{\text{cap}}(\theta_K, \theta_S) \\ & + \tilde{\beta}(\theta_K)\Big(1 - p_{\text{cap},\tau}\Big)\mathbb{E}\Big[V^K_{\tau+1}(\mu_{\tau+1})\Big] \end{aligned} \tag{S8.15}$$

The captor discount factor is set to $\tilde{\beta} = 0.92$ (default across all types). The baseline capture penalty under a Strict State is F_{cap} (DC: 40.704, PAR: 85.224, ELN: 55.968, FARC: 29.256). The reputational kill utility is η (DC: -3.760, PAR: -14.740, ELN: -5.530, FARC: -2.970). The release disutility is κ_{rel} (DC: 2.367, PAR: 14.273, ELN: 4.163, FARC: 1.471). The captor operational cost function is $C_\tau(\gamma) = \phi_C \exp(\kappa_c \gamma) + \nu_C$, where the cost parameters $(\phi_C, \kappa_c, \nu_C)$ are shown in Table S8.5:

Table S8.5: Captor Cost Coefficients $(\phi_C, \kappa_c, \nu_C)$

Captor Group	**Scale (ϕ_C)**	**Curvature (κ_c)**	**Shift (ν_C)**
DC	33.00	2.61	0.750
PAR	35.00	2.63	0.500
ELN	38.00	2.70	0.250
FARC	40.00	2.70	0.000